

Jason Landa

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Education

California State University, Long Beach

Expected Graduation: *December 2026*

B.S. Aerospace Engineering, Astronautics Emphasis

Relevant Coursework: Computational Fluid Dynamics, Thermodynamics, Heat Transfer System Design, Aerodynamics I-II, Propulsion, Aerospace Structures I-II, Fluid Dynamics, Fundamentals of Spacecraft Dynamics, Mechanics of Deformable Bodies,

Software & Skills

Software: ANSYS, MATLAB, SolidWorks, Excel, Python, Mastercam, CATIA V5

Technical Skills: CFD (steady/transient turbulent flow, heat transfer), Rocket Propulsion, Workshop Instruction, CAD, FEA, CAM Toolpath Generation, Trade Study Optimization, Lathe, Mill, Wire EDM, Soldering, Smoke-Wire Visualization, Schlieren Imaging, Wind Tunnels, Pitot-Static Probes, Pressure Transducers, Hot-Wire Anemometers

Experience

Lunar Lander Design Team (AURORA) – Propulsion design Lead (CSULB)

Jan 2026 – Present

- Led propulsion subsystem to develop design iterations of a LOX/RP-1 regenerative cooled rocket engine, throttleable pintle injector and cold gas thruster assembly. Assigned members to conduct gas dynamics simulations, thermomechanical stress analysis, pintle spray angle simulations, combustion stability analysis, and design optimization studies.
- Conducted parametric design sweeps across multiple pintle injector performance parameters using MATLAB-based cost function analysis to identify the optimal throttleable geometric pintle configuration, advancing the pintle injector through Preliminary Design Review (PDR).
- Conducted structural verification of regeneratively cooled rocket engine components using two-way fluid–structure interaction (FSI) simulations in ANSYS Fluent and ANSYS Mechanical to evaluate thermal stress distributions, deformation behavior, and structural integrity under coupled fluid and thermal loading conditions.
- Developed SHOCKWVE software, a nozzle generating GUI to rapidly generate and export nozzle contours for CFD and thermal simulations, streamlining supersonic nozzle iteration and analysis workflows.
- Modeled Steady-State heat transfer cooling channel design by conducting 1D numerical preliminary studies for the (BOREALIS) regenerative cooled rocket engine using python, supporting subsystem validation and advancing the engine through Preliminary Design Review (PDR).

Beach Launch Team (BLT) – Pyrotechnic Lead, ANSYS Fluent Instructor (CSULB)

Jan 2025 – Present

- Conducted 1D and 2D CFD modeling of aerodynamic and propulsion-related flows for the LR101 regenerative cooled engine utilized in the BLT Theseus launch vehicle.
- Organized and facilitated ANSYS Fluent training sessions focused on advanced CFD simulation methodologies and best practices.
- Conducted regenerative cooling performance studies using calibrated infrared thermography, mapping temperature fields across 300+ data points and improving thermal prediction correlation by 20%.
- Executed hydrodynamic flow testing to calculate discharge coefficients across combustion chamber geometries, reducing flow uncertainty by 12% and informing inlet and outlet design modifications.
- Directed repeatability and performance testing of pyrotechnic ignition materials, analyzing ignition success rates and thermal response metrics to guide primary igniter selection.
- Redesigned the pyrotechnic igniter assembly including firing circuit revisions, mechanical retention features, and initiation charge containment to improve structural robustness under vibration and thermal loads.
- Manufactured igniter bracket assemblies using manual lathe machining techniques and precision fabrication methods to ensure proper fitment, structural integrity, and repeatable integration within the ignition system assembly.

Gas Dynamics Research Assistant – Compressible Flow Optimization (CSULB)

Sep 2025 – Present

- Developed two-phase vaporization and heat-transfer models that reduced prediction error by 15%, improving chamber pressure and temperature correlation for chamber design decisions.
- Developed MATLAB and Python-based Prandtl–Meyer Method of Characteristics models to design and optimize a shock-free supersonic nozzle, achieving less than 5% flow-angle error and improving expansion efficiency.
- Performed comprehensive chemical-equilibrium analysis (CEA) trade studies across candidate propellant blends, generating contour and sensitivity plots to map specific impulse versus mixture ratio and recommend optimal propellant combinations.
- Developed a nozzle optimization methodology accounting for heat-transfer losses within the flow field to preserve target thrust performance by iteratively modifying nozzle geometry to compensate for thermally induced thrust degradation.